REAL-TIME ATHLETE PERFORMANCE ANALYTICS AND ANOMALY DETECTION

Project Report — Draft | Mohamed Riyaz S. | VIT Chennai

Abstract

This project presents an educational athlete sensor analytics pipeline that validates timestamped sensor records, engineers movement and exertion features, estimates a defined performance score using Random Forest regression, detects unusual sensor patterns using Isolation Forest, and generates transparent rule-based alerts. A Streamlit dashboard is provided to explore sensor trends, predicted scores, anomaly flags, and alert records. The bundled generator produces synthetic data solely to demonstrate the workflow; synthetic outputs are not evidence about real athletes.

1. Introduction

Wearable and training sensors can produce continuous streams of heart-rate, speed, and acceleration measurements. Transforming these measurements into understandable summaries can support training analysis and performance monitoring. This project demonstrates a modular analytics workflow from sensor ingestion through feature engineering, prediction, anomaly detection, alerts, and dashboard visualization.

2. Problem Statement

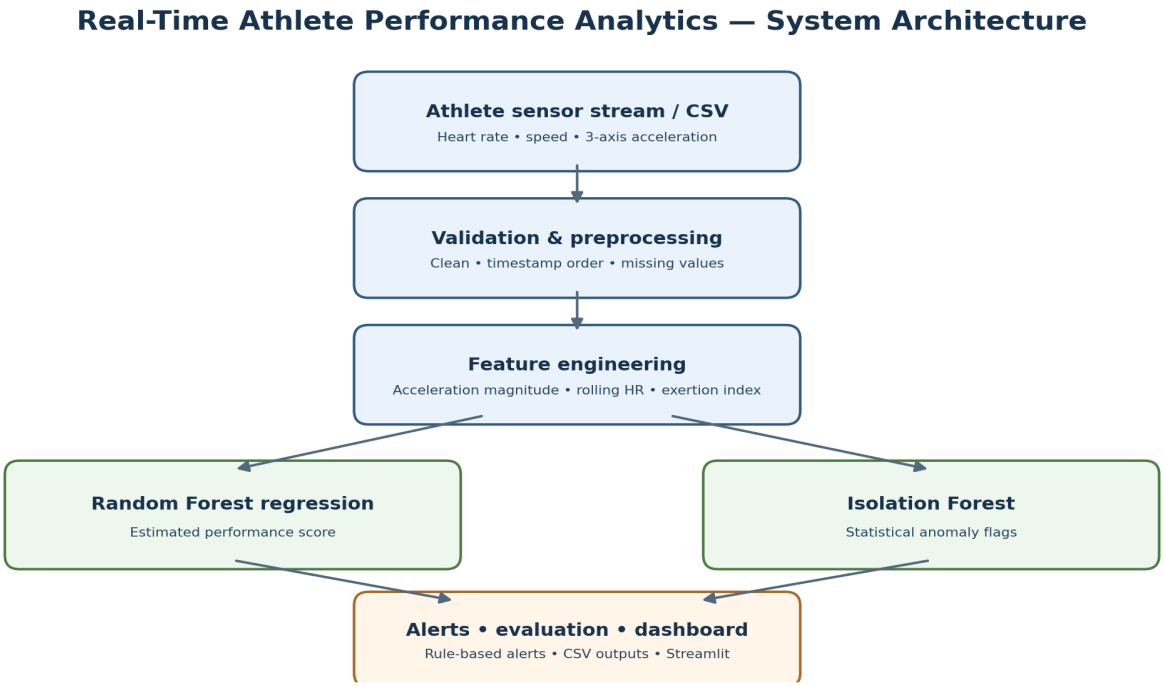
Raw athlete sensor readings are difficult to inspect manually at scale. Analysts need a reproducible way to summarize sensor patterns, estimate a clearly defined performance outcome, flag unusual observations for review, and communicate findings. This prototype addresses the software workflow while recognizing that real-world validity depends on authorized data, validated targets, domain-reviewed thresholds, and independent evaluation.

3. Objectives

- Build a reproducible sensor-data preprocessing pipeline.
- Engineer acceleration, rolling heart-rate, change, normalized, and exertion features.
- Train a Random Forest regression model for a defined performance-score target.
- Apply Isolation Forest to identify statistically unusual sensor patterns.
- Generate transparent rule-based alerts and visualize outputs in Streamlit.

4. System Architecture

The proposed architecture is shown below. The implementation currently processes CSV data and simulates a sequential stream; it is not a production Kafka/MQTT streaming deployment.



5. Methodology

Sensor records are validated for required fields, timestamps, numeric values, duplicates, and ordering. Feature engineering derives acceleration magnitude, shifted rolling heart-rate mean and standard deviation, changes in heart rate and speed, acceleration change, normalized sensor features, and an exertion index. A group-based holdout is used for the Random Forest regression experiment so athlete IDs are separated between train and test. Isolation Forest is fitted to selected sensor features and returns statistical outlier flags. Rule-based alert conditions are configured separately from the model.

6. Evaluation

Regression evaluation uses MAE, RMSE, and R^2 against a defined target. Anomaly precision, recall, and F1 require ground-truth event labels; without labels, report anomaly rate and review events manually. For a real streaming deployment, also measure processing latency, throughput, dropped records, and alert delivery time. Do not treat a synthetic demonstration metric as real-world model validation.

7. Results

Insert results generated from the actual authorized dataset here. The included synthetic generator can validate code execution and dashboard wiring only. The demo performance_score is constructed for demonstration and must not be represented as a measured sports-science outcome.

8. Limitations and Responsible Use

Isolation Forest detects statistical deviations, not injuries, illness, fatigue diagnoses, or unsafe participation. Heart-rate and acceleration thresholds are device- and context-dependent. A performance-score proxy may encode arbitrary assumptions. This educational prototype must not be used for medical or athlete-safety decisions. Use informed consent, secure data handling, expert review, and independent validation.

9. Future Work

Future extensions include real sensor ingestion through a message broker, streaming windows, online feature computation, model drift monitoring, athlete-specific baselines, domain-validated alert thresholds, external validation, explainability, and deployment monitoring.

10. Conclusion

The project demonstrates an end-to-end software workflow combining sensor preprocessing, Random Forest performance estimation, Isolation Forest anomaly detection, rule-based alerts, and interactive visualization. Its real-world value must be established through consented sensor data, a defensible target definition, and independent validation.

References

Add the exact source citations for any real sensor dataset, sensor-device documentation, software libraries, and research papers used in the final version.